

Time-Bin BB84 QKD System Using Indium Phosphide and Silicon Nitride Photonic Integrated Circuits

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Abstract We demonstrate a dual-chip InP-SiN photonic QKD system with on-chip pulse generation and reconfigurable decoding, implementing time-bin BB84 with finite-key security against coherent attacks. The system sustains a QBER below 4 % and delivers secret keys at kbps rates over 150–250 km of fiber. © 2026 The Authors

Introduction

Unlike public-key cryptography, whose security is based on computational hardness assumptions increasingly challenged by quantum computing [1–4], quantum key distribution (QKD) derives its security from the fundamental laws of physics. The uncertainty principle and the no-cloning theorem guarantee that any eavesdropping on the key exchange produces detectable errors, thereby providing information-theoretic security against both present and future adversaries.

The practical deployment of QKD requires scalable and cost-effective hardware, making photonic integrated circuits (PICs) a compelling foundation [5]. PIC-based QKD has been demonstrated in silicon [6], silicon nitride (SiN) [7], indium phosphide (InP) [8], and lithium niobate [9], yet no single platform delivers active sources, high-speed modulation, and low-loss passive circuitry, motivating hybrid approaches that combine complementary strengths [10].

In this work, we present an InP-SiN dual-chip QKD system that implements the three-state time-bin BB84 protocol with one decoy state. Pairing active on-chip state generation in InP with reconfigurable low-loss decoding in SiN leverages the complementary strengths of both platforms, eliminates the long transmitter delay line required by interferometric time-bin schemes, and exposes a tunable basis-selection ratio for protocol optimization. The key rates are evaluated under a finite-key security analysis against coherent attacks [11], demonstrating secure operation across metropolitan- and intercity-scale fiber distances.

Experimental Setup

The experimental setup is illustrated in Fig. 1(a), with Alice and Bob PICs shown in Fig. 1(b) and Fig. 1(c), respectively, along with supporting electronics and detection equipment. Alice prepares weak coherent pulses on the InP PIC and

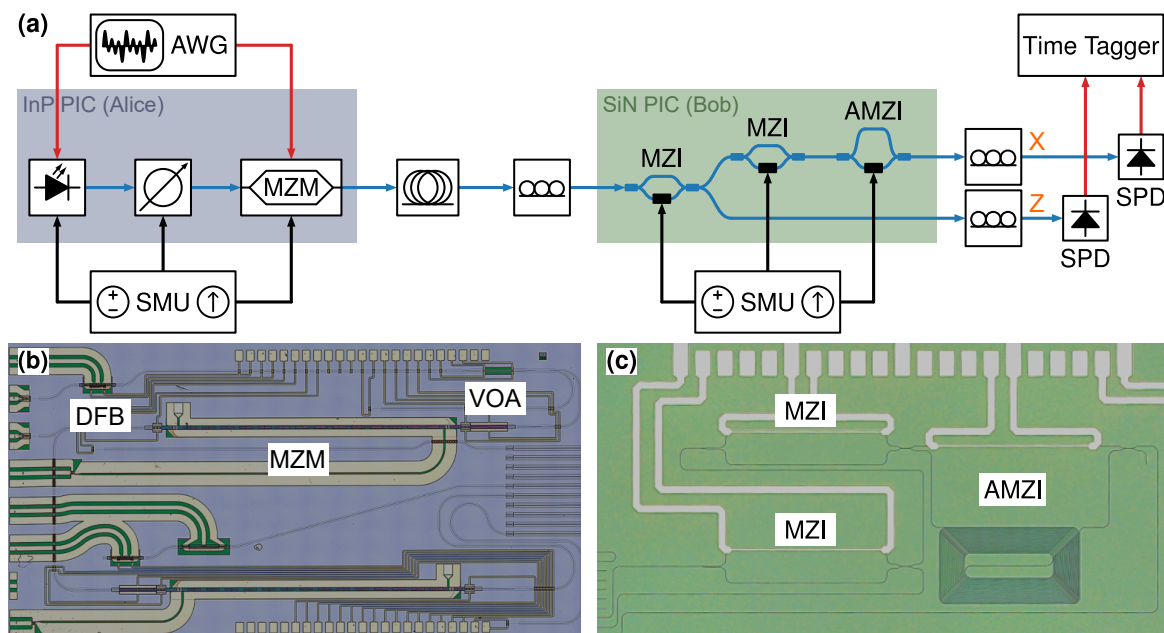


Fig. 1: (a) System-level schematic showing experimental setup. (b) Microscope image of the InP PIC (Alice). (c) Microscope image of the SiN PIC (Bob).

transmits to Bob over a spooled single-mode fiber link, where they are decoded by the reconfigurable SiN receiver and registered by superconducting nanowire single-photon detectors (SNSPDs).

The InP transmitter PIC comprises a directly modulated distributed feedback (DFB) laser, a variable optical attenuator (VOA), and a Mach-Zehnder modulator (MZM). Optical pulses are generated at a repetition rate of 1.065 GHz at 1550 nm, driven by an arbitrary-waveform generator (AWG) operating at 20 GS/s with 14-bit resolution. The DFB is gain-switched to produce phase-randomized pulses, the VOA provides coarse on-chip attenuation, and the MZM carves the time bins encoding the protocol states while setting the relative signal and decoy intensities, bringing the output down to the single-photon level. The transmitter PIC is biased with a multichannel source measure unit (SMU).

A polarization controller at the receiver input aligns the incoming polarization to the SiN PIC waveguide axis. The SiN receiver PIC [12] incorporates two Mach-Zehnder interferometer (MZI)-based tunable couplers and an asymmetric Mach-Zehnder interferometer (AMZI) with a delay line matching the 940 ps time-bin separation. Each tunable coupler is implemented as a symmetric MZI with a thermo-optic phase shifter (TOPS) in one arm, enabling precise control over the coupling ratio through thermo-optic phase tuning. The first tunable coupler sets the splitting ratio between the Z and X basis paths, while the second compensates for the loss imbalance introduced by the AMZI delay line, maximizing its extinction ratio. An additional TOPS in the short arm of the AMZI enables fine-tuning of the relative phase for X basis demodulation. All phase shifters are independently actuated by a second SMU.

Photons are detected by two channels of a Single Quantum Eos 800 CS SNSPD system, serving the Z and X basis outputs. The system offers 80 % system detection efficiency and 21 ps timing jitter at 1550 nm, with dark count rates below 10 cps per channel. Detection events are

timestamped by a Time Tagger, synchronized to the AWG via a shared reference clock; in deployment, this synchronization would be carried by a classical service channel or recovered from the quantum signal.

To enable phase-error estimation, the AMZI phase is tuned via its TOPS to produce destructive interference in the central slot of the X channel for the X_0 state. In a three-state protocol, Alice never sends X_1 , so any counts in that slot directly reflect phase errors, providing the most sensitive estimate of the phase error rate. The prepared quantum states are verified by recording time-bin histograms on the Time Tagger, as shown in Fig. 2.

Protocol Details

We implement the three-state time-bin BB84 protocol with one decoy state [13]. Alice randomly encodes one of two Z-basis states, Z_0 or Z_1 , as a pulse in the early or late time bin, or the X-basis state X_0 , an equal superposition of both time bins with zero relative phase. Each pulse is additionally assigned a random signal or decoy intensity [14]. Bob passively selects the measurement basis via the Z/X splitting ratio, which sets the basis-selection probabilities P_Z and $P_X = 1 - P_Z$ and can be adapted to channel conditions and decoy settings [15,16]. The photon arrival time in the Z path yields the raw key bit, while the central interference slot of the X path estimates the phase error rate bounding Eve's information.

These protocol requirements shape the chip-level design in two places: the transmitter time-bin generation and the receiver X-basis projection. At the transmitter, the 940 ps time bins are generated electronically by the AWG-driven MZM rather than through interference, avoiding a delay line on InP that would be prohibitive in both loss and footprint. At the receiver, the delay line required for X-basis projection is well within reach on SiN thanks to its low propagation loss, which also hosts the on-chip tunability discussed above and allows the protocol to be optimized across different operating regimes.

We estimate the extractable secret key length L following the finite-key security analysis of [11].

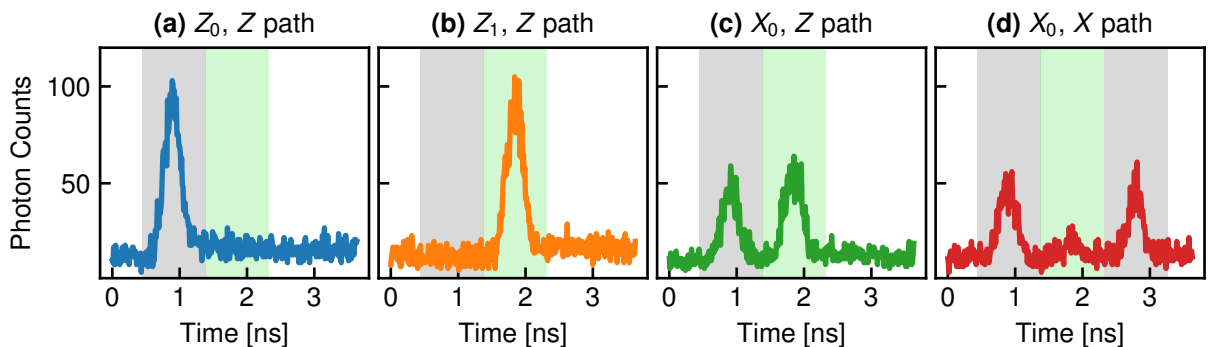


Fig. 2: Time-bin histograms of the states recorded at the receiver: (a) Z_0 and (b) Z_1 on the Z path, (c) X_0 on the Z path, and (d) X_0 on the X path, showing destructive interference in the central slot. The green shaded region indicates the active detection time-slot. Z-basis states on the X path are omitted as the single-pulse inputs do not produce interference.

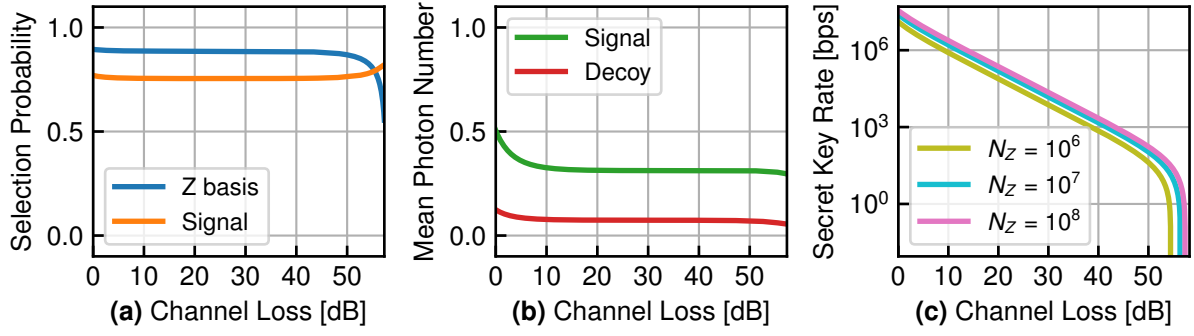


Fig. 3: Optimization and performance of the QKD system versus channel loss. **(a)** Optimal selection probabilities for the Z basis and the signal state. **(b)** Optimal mean photon numbers of the signal and decoy states. **(c)** Secret key rate for three values of the sifted Z-basis block size N_Z . Panels (a) and (b) are shown for a fixed $N_Z = 10^8$.

The key is distilled from the sifted Z-basis detections, with its secure length determined by the vacuum and single-photon yields, the phase error rate estimated in the X basis, and the information leaked during classical post-processing, giving

$$L = s_{Z_0}^L + s_{Z_1}^L (1 - h(\Lambda_X^u + \gamma)) - \text{leak}_{\text{EC}} - \log_2 \frac{2}{\epsilon_{\text{cor}}} - 4 \log_2 \frac{15}{\epsilon_{\text{sec}}' \sqrt{2}}. \quad (\dagger)$$

Here, $s_{Z_0}^L$ and $s_{Z_1}^L$ are lower bounds on the vacuum and single-photon contributions to the sifted key, obtained via the decoy-state method. The term Λ_X^u places an upper-bound on the single-photon quantum bit error rate (QBER) in the X basis, and γ is the Serfling correction [17] accounting for finite-sample estimation of the phase error rate. The error-correction leakage is $\text{leak}_{\text{EC}} = f_{\text{EC}} N_Z h(q_Z)$, with N_Z the number of sifted Z-basis detections, $h(\cdot)$ the binary entropy, q_Z the Z-basis QBER, and $f_{\text{EC}} = 1.16$ the reconciliation inefficiency. The secrecy and correctness parameters are $\epsilon_{\text{sec}}' = \epsilon_{\text{cor}} = 10^{-12}$.

Results

We evaluate the system performance in two steps. First, the protocol parameters are jointly optimized against the finite-key bound as a function of channel loss. Second, the QBER and extractable secret key rate (SKR) are estimated at loss values representative of metropolitan and long-haul deployment scenarios. Figures 3(a) and 3(b) show the loss-dependent values of the tunable parameters that maximize the SKR. The strong bias toward the Z basis is characteristic of the efficient-BB84 regime [18], where the Z basis dominates key generation while the X basis is used sparingly for phase-error estimation. The optimal values also depend on the sifted block size N_Z , since larger blocks tighten the statistical estimates and shift the optimal bias. Figure 3(c) accordingly shows the SKR for three values of N_Z , with the finite-size penalty shrinking and the secure distance extending as N_Z approaches the asymptotic regime.

We assessed the system at channel losses of 30 dB, 40 dB, and 50 dB, corresponding to 150 km, 200 km, and 250 km of standard single-mode fiber. The Z-basis QBER remained near 3% at the shorter distances and increased to just under 4% at 250 km, consistent with the reduced signal-to-noise margin at higher channel loss. The residual floor at low loss is set by a combination of state-preparation imperfections, finite AMZI extinction, and detector timing jitter leaking into adjacent slots. The corresponding estimated SKRs, evaluated at a sifted block size of $N_Z = 10^7$, are 16, 1.5, and 0.13 kbps, respectively.

Conclusions

We have reported a dual-chip InP–SiN photonic integrated QKD system implementing the three-state time-bin BB84 protocol with one decoy state, with its security assessed under a finite-key analysis against coherent attacks. The architecture exploits the strengths of the two platforms: pulse generation and time-bin carving in InP, and low-loss reconfigurable decoding with a tunable Z/X splitting ratio in SiN. The system sustained QBER values below 4% and delivered estimated SKRs of 16, 1.5, and 0.13 kbps over 150, 200, and 250 km of standard single-mode fiber. However, several directions remain open: tighter hybrid integration of active, modulation, and passive functionalities – together with low-loss co-packaging into a transceiver module, higher clock rates to boost raw key generation, the incorporation of compact on-chip single-photon detectors [19] to eliminate the cryogenic footprint of the current SNSPD stage, and active thermal and polarization stabilization for continuous unattended operation in the field. In parallel, quantifying source side-channels such as pattern-dependent leakage from the gain-switched DFB and residual intensity correlations between adjacent pulses [20] is an essential next step for certifying integrated transmitters. Together, these steps would bring chip-scale QKD closer to the compactness, cost, and scalability required for field deployment.

Acknowledgements

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